

Lakhdar Berache

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Exchange student at the Korea Advanced Institute of Science and Technology (KAIST, South Korea) and engineering student at the University of Technology of Troyes, passionate about software engineering and artificial intelligence. Strong experience built through academic, personal, and internship projects, and eager to keep developing my skills while contributing to innovative solutions and collaborative environments.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

Specialization in Informatics and Artificial Intelligence

Daejeon, South Korea

February 2025 – July 2025

- Exchange semester, courses taught in English.
- Academic projects in Data Science, Artificial Intelligence, Machine Learning, and Start-up Management.

University of Technology of Troyes (UTT)

Specialization in Informatics and Systems

Troyes, France

2024 – 2027

- Integrated Preparatory Program (Tronc Commun, 2022 – 2024).
- Academic projects in database management and software engineering.

WORK EXPERIENCE

ArcherMind Technology (Nanjing) Co., Ltd.

Consultant Software & AI Engineer

Paris, France

March 2026 – Present

- Built and improved AI agents for client projects, translating business needs into reliable software solutions.
- Conducted technical analysis and produced clear software documentation to support product and development decisions.

Renault Korea

Software & AI Engineer Intern

Yongin, South Korea

September 2025 – February 2026

- Developed a connected health & fitness application with AI-powered personalized recommendations and computer-vision analysis.
- Co-developed an AI-driven podcast generation platform using NLP and TTS, producing fully personalized content from user preferences.
- Built a multi-agent conversational AI system with LangGraph, integrating MCP servers, a RAG pipeline for vehicle-manual knowledge retrieval, real-time TTS streaming, and cloud messaging for mobile notifications.
- Presented monthly AI project progress and strategic insights to directors and the CEO, supporting data-driven decision-making.

EXTRACURRICULAR ACTIVITIES

Rebloom

Founder & CEO

Paris, France

Ongoing

- Built an anti-flower-waste marketplace connecting florists with customers around discounted unsold bouquets.
- Implemented end-to-end payment processing and designed a scalable, multi-platform architecture serving daily active users.
- Pitched Rebloom at startup competitions and investor events, building the pitch deck, business model, and go-to-market strategy.

CarChat

Personal Project

Troyes, France

Ongoing

- Built an AI assistant for vehicle owner manuals delivering instant, source-backed answers and reducing documentation search time.
- Implemented a RAG workflow with LangChain to index manuals by car model and retrieve the most relevant sections, improving answer accuracy.

Megawatt'UTT Challenge by EDF

National Engineering Competition

Troyes, France

June 2026

- Won 2nd place and the Public's Prize, judged on risk-taking, feasibility, and implementation quality.
- Developed a mobile training simulator for industrial crane operation, combining real-time 3D simulation with a web-based learning and progress-tracking platform.
- Reproduced the industrial HMI (Magelis / SOMDEL screen); featured in *Le Parisien* with a dedicated article on the team and prototype.

STUDENT PROJECTS

RePLY – Intelligent Heart Rate Monitor

Daejeon, South Korea

Machine Learning / Wearable

June 2025

- Developed an intelligent wearable device using Machine Learning to monitor heart rate, estimate calorie expenditure, and optimize exercise through haptic feedback.

AI Adventure

Daejeon, South Korea

Learning Platform Concept

June 2025

- Designed a student-focused learning platform that turns research-paper databases into accessible summaries, learning guidance, and source-based study support.
- Framed the product around a startup innovation approach (Y Combinator template) to deliver a student-centered solution.

ECU Car Testing

Troyes, France

Software Testing

May 2024

- Built a software-testing app showing how a vehicle control system should react in normal driving and failure conditions, with clear pass/fail results.
- Added fault simulation and safety checks to demonstrate automotive reliability-testing concepts, following the Expected vs Actual rule (test oracle).

Scout Mini

Troyes, France

Robotics / ROS2

May 2024

- Co-developed a comprehensive user guide for the AgileX Scout Mini robot.
- Leveraged ROS2 for robot operation and worked on implementing automation workflows.

SKILLS & INTERESTS

Languages: French (native), Arabic (fluent), English (fluent), Korean (beginner).

Programming: Python (OOP), Pandas, Matplotlib, Scikit-learn, LangChain, Hugging Face Transformers.

Front-End: React.js, JavaScript • **Back-End:** Flask, FastAPI • **Databases:** MySQL, MongoDB, Supabase.

AI & Machine Learning: NLP, LLMs, Computer Vision (YuNet, ResNet) • **Robotics:** ROS2.

Engineering & PM: Agile, UML, Gantt charts • **Soft skills:** Communication, Curiosity, Adaptability, Organization.

Certifications: Generative AI (Microsoft), Computer Vision App with Azure (Microsoft), Build a RAG Application (IBM).

Interests: Technology & AI, Sport (Football, Volleyball), Art, Entrepreneurship.